Ryan Wien

Fintech portfolio - six working systems, live demos included

Live site + demos: ryanwien.github.io/FintechPortfolio
Source: github.com/ryanwien/FintechPortfolio
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Six production-style systems covering the jobs fintech companies hire for: moving money, catching fraud, approving loans, running markets, and compliance. 413 automated tests pass on every change across 18,542 lines of Python. Every number below is produced by the code itself, each project states its own limitations, and each one has an interactive demo on the site.

Double-entry ledger & payments API

Payments infrastructure · 113 tests

The bookkeeping engine behind a payments app: moves money between accounts without ever losing a cent, even when the same request is sent twice or a thousand arrive at once.

14.752 ms p99 transfer latency | 556 transfers/s transfer throughput | 100% precision recon break classification

Real-time card fraud scoring

Risk & payments ML · 52 tests

Decides in under a millisecond whether a card payment looks stolen, and prices the decision in dollars: block too little and fraud walks, block too much and real customers get declined.

0.884 PR-AUC (held-out months) | 33.6 % Recall @ 0.5% alert rate | 50,160 USD/month Expected loss saved vs rules

Credit underwriting scorecard with model-risk documentation

Lending & model risk · 65 tests

Decides who qualifies for a loan the way a regulated lender has to: provably fair across groups, with the exact reasons every declined applicant is legally owed.

0.700 OOT AUC (champion) | 0.017 Score PSI dev vs OOT | 0.914 Adverse impact ratio

Limit order book matching engine + event-driven backtester

Trading & market infrastructure · 45 tests

A miniature stock exchange: matches buyers with sellers at 150,000+ orders per second, the core of how markets like NASDAQ work.

214,338 orders/s Engine throughput | 17.3 us p99 per-order latency | identical Replay determinism

C++ matching engine: the Python book, compiled

Trading & market infrastructure · 105 tests

The matching engine rebuilt in C++20: the same order book, rules and tests, running an order of magnitude faster on the same machine.

4,321,213 orders/s Engine throughput | 0.60 us p99 per-order latency | 20.2x Speedup vs Python twin

AML Transaction Monitoring

Financial crime & compliance · 33 tests

The compliance watchdog banks are required to run: spots the patterns criminals use to launder money and keeps a tamper-proof evidence trail for regulators.

87.8 % Typology detection at the line | 54.6 % Alert precision | 3.67 % False-positive rate

Stack: Python · FastAPI · scikit-learn · pandas · SQLite · networkx · Hypothesis · pytest · GitHub Actions (CI on Python 3.12-3.14). Money is never a float; no lookahead; deterministic synthetic data, so everything reruns with one command.